				[2]		
10			**DR** $\frac{dy}{dx} = \frac{4}{\pi} - \frac{8x}{\pi^2} - \cos x$	**M1**	**3.1a**	Attempt to differentiate. Both an x term and a cos term needed condone other errors. If π^2 is treated as a variable and incorrectly differentiated eg to 2π then M0. Similarly with π.
			$x = 2.6$, $\frac{dy}{dx} = 0.0226...$	**M1**	**3.1a**	At least one substitution into their expression
			$x = 2.7$, $\frac{dy}{dx} = -0.0112...$	**A1**	**1.1**	Both correct (at least 2d.p., rounded or truncated)
			Gradient zero for a value between 2.6 and 2.7	**E1**	**2.2a**	Can be implied by 'sign change' or sketch Dependent on M2 but can be earned following **M2A0**
			Gradient positive, zero, negative so max	**B1**	**2.4**	Allow 2nd derivative used at turning point found BC (x= 2.67, 2nd deriv = -0.356) or 2 relevant values eg -0.295 and -0.383 seen and stated as < 0 therefore max
				[5]		